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Towards Inclusive Mobility Modeling: Characterizing elderly trajectory patterns in urban systems

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Background: Modern Intelligent Systems Depend on Data



LLMs & Analytics

Text and interaction data

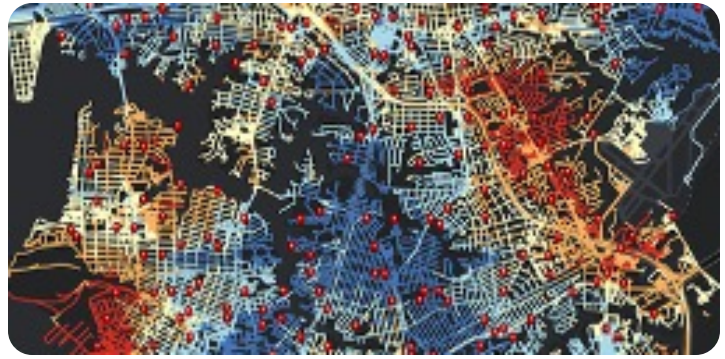
Learn from recorded language and observations



Autonomous Driving

Road and sensor data

Predict from previously observed scenarios



Urban Intelligence

Mobility and population traces

Turn activity records into planning decisions

They share one principle in common: models can only learn from the data we provide.

Motivation: Older Adults Are Less Visible in Digital Data

Two empirical data highlight the phenomenon of unequal age representation.

Web Usage

2024

Online time > 10h

Ages 18–29		Ages 65+
62%	vs.	15%

Generative AI Usage

2025

Have ever used ChatGPT

Ages 18–29		Ages 65+
58%	vs.	10%

WHO Warning

"The data used by AI can be unrepresentative of older people or skewed by past ageist stereotypes, prejudice or discrimination."

Lower participation can become lower representation in training data—and ultimately in model outputs.

1. Pew Research Center. Americans' Use of Mobile Technology and Home Broadband. 2024.
2. Pew Research Center. ChatGPT Use Among Americans Roughly Doubled Since 2023. 2025.
3. World Health Organization (WHO). Ageism in Artificial Intelligence for Health. 2022.

Research Question

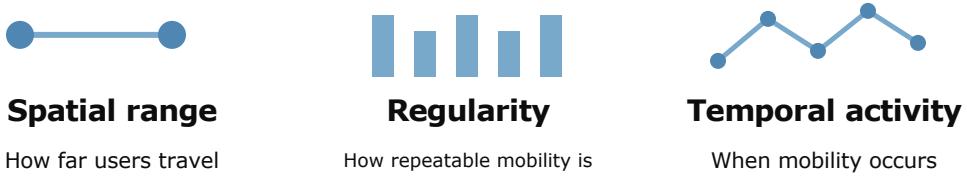
MAIN RESEARCH QUESTION

How does the underrepresentation of a minority group affect downstream tasks?

Around this main RQ, we discuss the following two aspects:

1. Age-specific mobility characterization

Compare older and younger users across three behavioral dimensions.



Spatial range
How far users travel

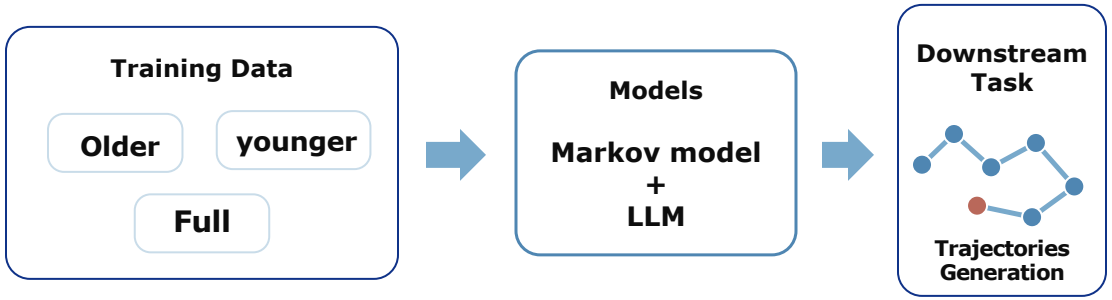
Regularity
How repeatable mobility is

Temporal activity
When mobility occurs

Older adults vs. younger users

2. Controlled downstream experiment

Change population composition while keeping the downstream task fixed.



Compare the same downstream tasks under different population compositions.

Data Source: CitiBike Offers a Longitudinal View of Urban Mobility



System footprint: Jersey City

Each line connects a published start and end station.

We use data in Jersey City from 2016–2020

Trip-level records include stations, coordinates, duration, user type, birth year, and gender.

1,638,153	60	93.3%
raw trips	months	records retained

Quality-controlled dataset

Removed records with Haversine distance > 10 km or computed speed > 10 m/s.

Age-defined subsets

Elderly (65+)	14,394 trips	**0.9%**
Young (18–35)	783,558 trips	**51.5%**
Others	1,638,153 trips	**47.6%**

Descriptive Statistic

Definition:

One trajectory
3 trip records



Young vs. elderly riders

Bars normalized to the young group — Young — Elderly

Trip records	Elderly = 1.8% of young	**783,558** / **14,394**
Trajectories	Elderly = 34.4% of young	**5,832** / **2,007**
Trips per trajectory	18.7× lower for elderly	**134.4** / **7.2**
Unique stations	Elderly = 90.9% of young	**55** / **50**

Elderly mobility shows lower overall mobility intensity but comparable spatial coverage.

Defining Other Six Metrics to Characterize Mobility Patterns

Spatial Dimension

Where and how broadly riders move



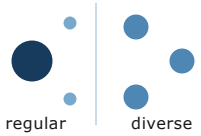
Step length

Straight-line distance from origin to destination.



Radius of gyration (RoG)

Spread of visited stations around the activity center.



Mobility entropy

Diversity of station visits; lower means more regular.

Temporal Dimension

How quickly and when mobility occurs



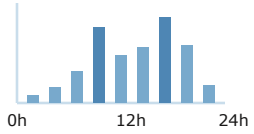
Speed

Distance traveled per unit time.



Dwell time

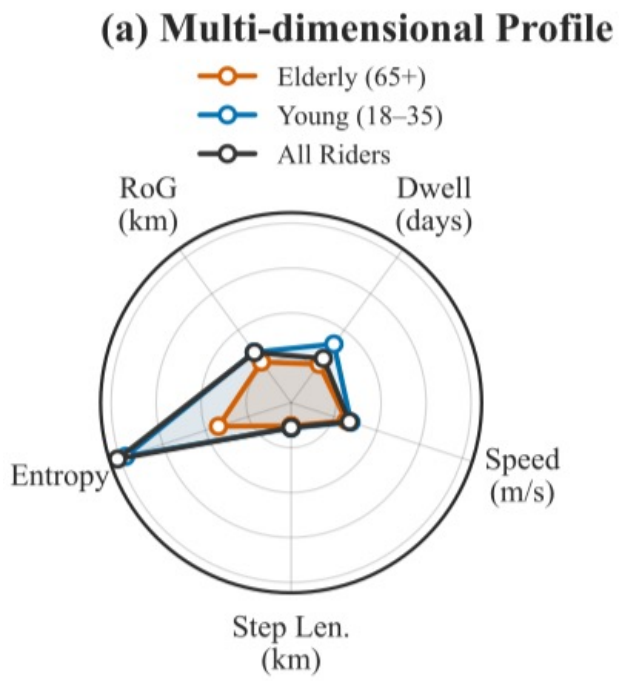
Age-group-specific Bike-use Recurrence Interval

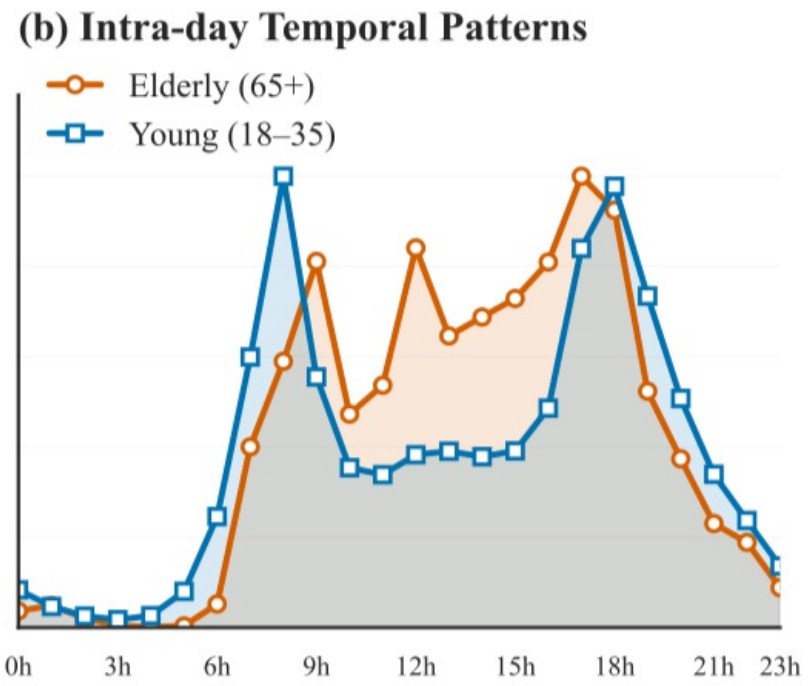


Intra-day activity

Distribution of trip starts across the 24-hour day.

Older and Younger Riders Follow Distinct Mobility Rhythms





More regular, more concentrated

Older riders show lower entropy and a slightly smaller radius of gyration.

A different daily rhythm

Young riders have sharper commute peaks; older activity remains broader through midday.

The Tested Downstream Task: Trajectory Generation

We conduct a 3 × 2 Controlled Design Isolates Data and Model Effects

Training Population

LLM fine-tuning corpora after split

A	**All riders** majority-dominated	**2,984 trajectories** 667,013 trips
Y	**Young only** ages 18–35	**2,897 trajectories** 469,801 trips
E	**Elderly only** ages 65+	**1,631 trajectories** 13,928 trips

Two Model Paradigm

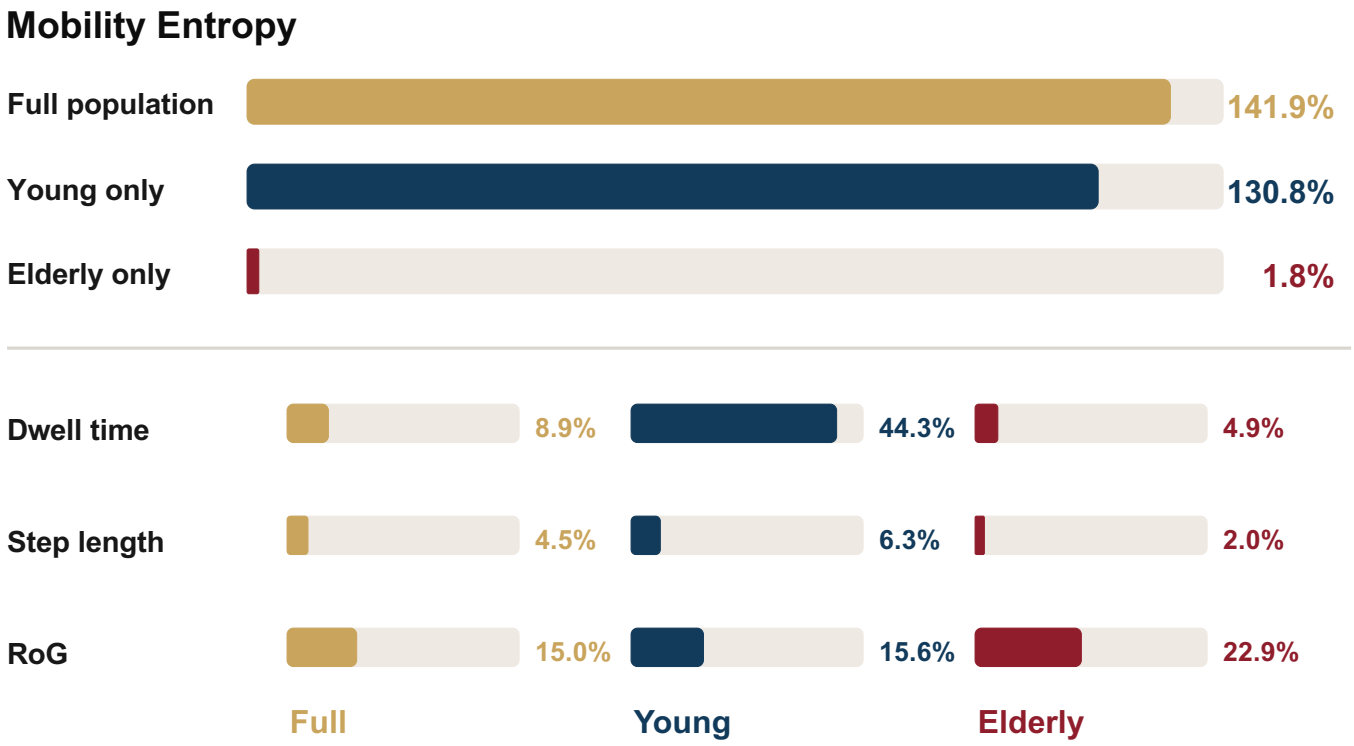
	First-order Markov	Retraining Qwen3-4B
ALL RIDERS	M_{all}	L_{all}
YOUNG ONLY	M_{young}	L_{young}
ELDERLY ONLY	$M_{elderly}$	$L_{elderly}$

VS

whether a more powerful foundation model can compensate for the limited data available for minority group.

Markov Results

Absolute Relative Error. Smaller is better.



Key findings

01 Representation matters

$M_{elderly}$ reduces Step Length, Dwell Time, and Entropy errors to below 5%.

02 Coverage still matters

Its RoG remains 22.9% lower than the real elderly baseline because elderly transitions are sparse.

Full population data cannot recover elderly spatial behavior when the majority population dominates.

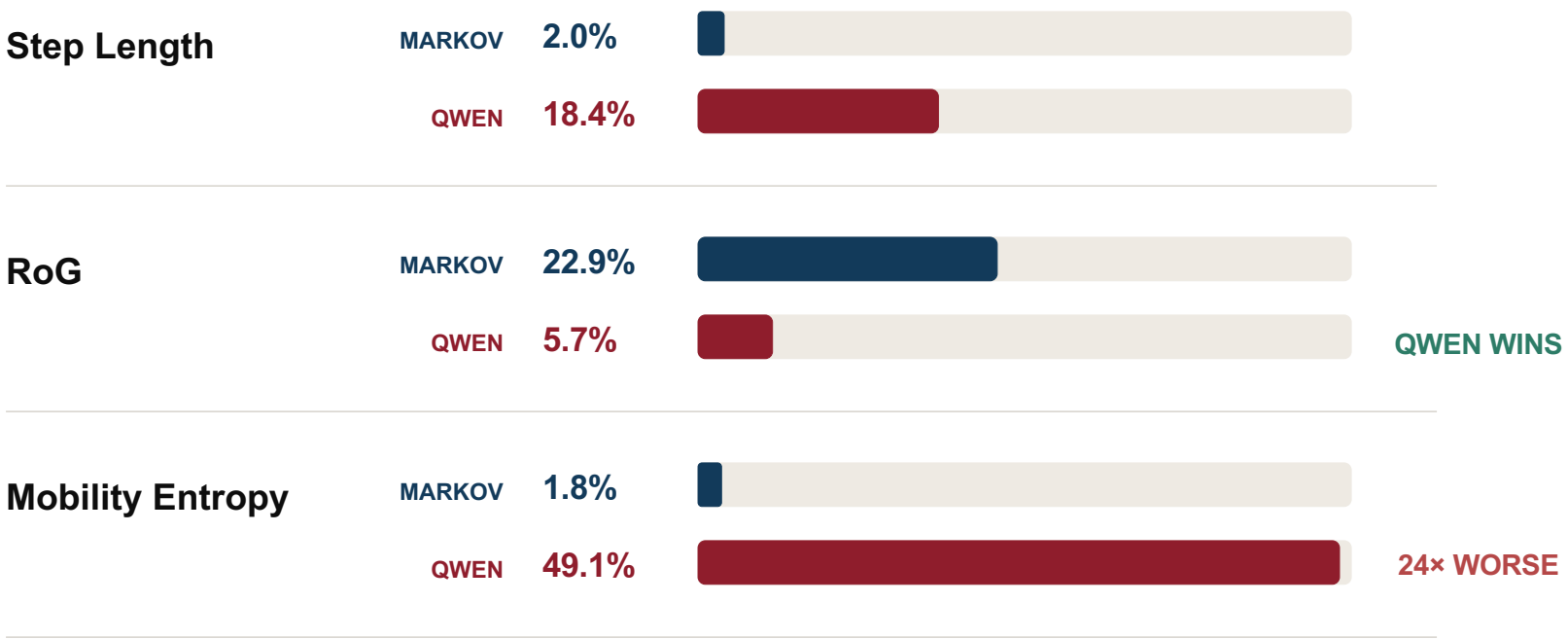
LLM Results

Table 4. LLM-generated synthetic trajectories vs. Real Elderly. Bold indicates best among the three LLM variants.

Metric	Real Elderly	$\mathcal{L}_{\text{elderly}}$	$\mathcal{L}_{\text{young}}$	$\mathcal{L}_{\text{all}}$
Step Length (m)	1,004.3	1,189.2 (+18.4%)	967.3 (−3.7%)	**878.8 (−12.5%)**
Speed (m/s)	2.41	4.15 (+72.3%)	4.15 (+72.2%)	4.16 (+72.4%)
Dwell Time (s)	19,338	597 (−96.9%)	601 (−96.9%)	605 (−96.9%)
RoG (m)	957.8	**903.6 (−5.7%)**	846.9 (−11.6%)	773.5 (−19.2%)
Entropy	1.82	**2.72 (+49.1%)**	3.06 (+67.6%)	2.86 (+57.0%)

Elderly-specific still performs poorly in capturing mobility diversity and temporal patterns.

Markov vs Qwen3-4B



Spatial coverage
RoG error: 22.9% → 5.7%

Behavioral regularity
Entropy error 1.8% → 49.1%

Takeaways:
1. **Markov is more faithful overall. Win 4 in 5 metrics.**
2. **Greater model complexity does not guarantee greater data fidelity**

Conclusion

Takeaways:

01 **Mobility patterns are structurally different**

Elderly and young riders differ across spatial range, mobility diversity, and temporal activity.

02 **Majority data creates subgroup bias**

Models trained on majority-dominated populations systematically misrepresent elderly mobility behavior.

03 **Higher capacity model is not a solution**

LLM-based generation does not guarantee better subgroup fidelity when demographic data remain sparse.

Thanks for listening!

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